

The schematic diagram illustrates the RP2040 hardware design, centered around the RP2040 microcontroller (U2). The microcontroller is connected to a power supply network consisting of a +3V3 rail and a +1V1 rail. The +3V3 rail is connected to the VREG_VDD pin (pin 44) and the VREG_VDD pin (pin 43). The +1V1 rail is connected to the VREG_VDD pin (pin 45) and the VREG_VDD pin (pin 46). The microcontroller is also connected to a USB interface (USB_DP and USB_DM) and a QSPI interface (QSPI_SS, QSPI_SD0, QSPI_SD1, QSPI_SD2, QSPI_SD3, QSPI_SCLK). The microcontroller is connected to a KBD interface (KBD_SDA and KBD_SCL) and a BL_CONTROL pin (pin 17). The microcontroller is connected to a SWCLK pin (pin 24) and a SWD pin (pin 25). The microcontroller is connected to a TESTEN pin (pin 19) and a GND pin (pin 57). The microcontroller is connected to a PWR_FLAG pin (pin 41) and a PWR_FLAG pin (pin 42). The microcontroller is connected to a PWR_FLAG pin (pin 43) and a PWR_FLAG pin (pin 44). The microcontroller is connected to a PWR_FLAG pin (pin 45) and a PWR_FLAG pin (pin 46). The microcontroller is connected to a PWR_FLAG pin (pin 47) and a PWR_FLAG pin (pin 48). The microcontroller is connected to a PWR_FLAG pin (pin 49) and a PWR_FLAG pin (pin 50). The microcontroller is connected to a PWR_FLAG pin (pin 51) and a PWR_FLAG pin (pin 52). The microcontroller is connected to a PWR_FLAG pin (pin 53) and a PWR_FLAG pin (pin 54). The microcontroller is connected to a PWR_FLAG pin (pin 55) and a PWR_FLAG pin (pin 56). The microcontroller is connected to a PWR_FLAG pin (pin 57) and a PWR_FLAG pin (pin 58). The microcontroller is connected to a PWR_FLAG pin (pin 59) and a PWR_FLAG pin (pin 60). The microcontroller is connected to a PWR_FLAG pin (pin 61) and a PWR_FLAG pin (pin 62). The microcontroller is connected to a PWR_FLAG pin (pin 63) and a PWR_FLAG pin (pin 64). The microcontroller is connected to a PWR_FLAG pin (pin 65) and a PWR_FLAG pin (pin 66). The microcontroller is connected to a PWR_FLAG pin (pin 67) and a PWR_FLAG pin (pin 68). The microcontroller is connected to a PWR_FLAG pin (pin 69) and a PWR_FLAG pin (pin 70). The microcontroller is connected to a PWR_FLAG pin (pin 71) and a PWR_FLAG pin (pin 72). The microcontroller is connected to a PWR_FLAG pin (pin 73) and a PWR_FLAG pin (pin 74). The microcontroller is connected to a PWR_FLAG pin (pin 75) and a PWR_FLAG pin (pin 76). The microcontroller is connected to a PWR_FLAG pin (pin 77) and a PWR_FLAG pin (pin 78). The microcontroller is connected to a PWR_FLAG pin (pin 79) and a PWR_FLAG pin (pin 80). The microcontroller is connected to a PWR_FLAG pin (pin 81) and a PWR_FLAG pin (pin 82). The microcontroller is connected to a PWR_FLAG pin (pin 83) and a PWR_FLAG pin (pin 84). The microcontroller is connected to a PWR_FLAG pin (pin 85) and a PWR_FLAG pin (pin 86). The microcontroller is connected to a PWR_FLAG pin (pin 87) and a PWR_FLAG pin (pin 88). The microcontroller is connected to a PWR_FLAG pin (pin 89) and a PWR_FLAG pin (pin 90). The microcontroller is connected to a PWR_FLAG pin (pin 91) and a PWR_FLAG pin (pin 92). The microcontroller is connected to a PWR_FLAG pin (pin 93) and a PWR_FLAG pin (pin 94). The microcontroller is connected to a PWR_FLAG pin (pin 95) and a PWR_FLAG pin (pin 96). The microcontroller is connected to a PWR_FLAG pin (pin 97) and a PWR_FLAG pin (pin 98). The microcontroller is connected to a PWR_FLAG pin (pin 99) and a PWR_FLAG pin (pin 100).

[illegible]